Probably Approximately Correct String Theory: A PAC Verification Framework for the String Landscape

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Abstract

String theory's landscape contains on the order of 10^{500} vacua and no experiment distinguishes most of them, so "unified string theory" conflates two problems—*verification* (does some vacuum match our universe?) and *selection* (why that one?). We separate them. Importing Leslie Valiant's computational learning framework, we make verification precise as a *probably-approximately-correct certificate*: with probability $\geq 1-\delta$ a candidate vacuum's physics agrees with observation within tolerance ε, *even when the verifier is itself approximate*—a two-level bound we derive. We show single-candidate verification needs *no learnability assumption*, while *search* over the landscape is governed by Valiant's evolvability theorem (*evolvable* $\Rightarrow$ *learnable*), yielding one necessary-condition conjecture: the compactification$\to$observables map is SQ-learnable. Finally, making the observable measure finite collapses the sampling term and exposes the binding constraint—verification is bounded by the number of *measurable* points of the true theory, ~ 19 for the Standard Model, which today gives a certificate radius of $\sim 43\%$, not $\sim 1\%$. The framework's product is not a solution to unification but a *quantitative statement of the verification deficit*—and a precise sense in which string theory is, today, barely verifiable.

1 Introduction

String theory is not, in its present form, a single theory. It is a family of $\sim 10^{500}$ metastable vacua—the *landscape*—each giving different low-energy physics [1, 2, 3]. The statement "string theory is unified" decomposes into two claims that the literature habitually fuses:

1. **Verification**—there exists a vacuum whose low-energy physics matches our universe (gauge group, spectrum, cosmological constant).
2. **Selection**—a *measure* singles out that vacuum, with weight ~ 1.

The first is checkable in principle; the second is not, and it is the part that has remained open for roughly four decades. This paper concerns only the first, and does so rigorously.

Two facts make verification hard even as a well-posed problem. First, the search space is astronomically large. Second, and more fundamentally, the *verifier* is not ground truth: the map from a compactification to its low-energy observables is computed by an approximate solver—with systematic error from α' corrections, non-perturbative truncations, and the Kaluza–Klein tower. Any rigorous statement about verification must therefore contend with an *imperfect labeler*.

We import the one framework built for exactly this regime. Valiant's probably-approximately-correct (PAC) learning replaces "exact" with "with probability $\geq 1-\delta$, error $\leq \varepsilon$" [4]. We apply that replacement to verification itself, and derive a certificate whose two error terms are separately estimable. Three consequences follow: a clean separation of *verification* (no learnability needed) from *search* (governed by Valiant's evolvability theorem [5]); a necessary-condition conjecture for the whole landscape program; and a bound that shows verification today is limited not by compute but by *measurable physics*.

1.1 Contributions

1. **A two-level certificate** (Theorem 3): an upper bound on the true error of a candidate vacuum that composes a candidate-vs-oracle term with an oracle-vs-truth term, valid when the verifier is approximate.

2. **The verify/search separation** (Section 4): single-candidate verification needs no VC dimension; search does, and the framework degrades gracefully when learnability fails.

3. **The measurable-physics bound** (Section 5): a finite observable measure collapses the sampling term, leaving a certificate radius set by the number of *measured* points—$\sim 43\%$ today, $\sim 1\%$ only with $\sim 18{,}000$.

4. **The learnability conjecture** (Section 6): the compactification→observables map is SQ-learnable; its false branch is a negative result about the entire landscape program.

2 Setup and notation

Let X be the *instance space*—the set of low-energy observables (masses, couplings, the cosmological constant). Let D be a probability measure over X: *which* observables are sampled, and with what weight. D is a stipulated scientific commitment, not given by the physics (Sections 5 and 7).

Let $c^* : X \to Y$ be the *target*—the true low-energy physics of our universe—and let $h : X \to Y$ be a candidate vacuum's predicted physics. We never observe c^* directly; we observe it through an *oracle* $\widehat{O} : X \to Y$, an approximate solver. Define the two disagreement functions

$$g(x) \;=\; |h(x) - \widehat{O}(x)|, \qquad e(x) \;=\; |\widehat{O}(x) - c^*(x)|,$$

and the *true error*

$$\mathrm{err}_D(h) \;=\; \Pr_{x\sim D}\big[\, |h(x) - c^*(x)| > \varepsilon \,\big]$$

for a chosen tolerance ε. The goal is to certify $\mathrm{err}_D(h) \le \delta$ for small δ. We cannot observe $\mathrm{err}_D(h)$, because c^* is reachable only through $\widehat{O}$.

3 The two-level certificate

The triangle inequality decomposes the unobservable error into two estimable pieces:

$$|h(x) - c^*(x)| \;\le\; g(x) + e(x),$$

hence, for any split point $\beta \in (0, \varepsilon)$,

$$\{\, |h - c^*| > \varepsilon\} \;\subseteq\; \{g > \varepsilon - \beta\} \;\cup\; \{e > \beta\},$$

and, taking probabilities (a union bound),

$$\mathrm{err}_D(h) \;\le\; p_g(\varepsilon - \beta) + p_e(\beta), \tag{1}$$

where $p_g(\tau) = \Pr[g > \tau]$ and $p_e(\beta) = \Pr[e > \beta]$. The two terms on the right are *individually estimable*, for different reasons: p_g compares two things we can both *compute* (h and $\widehat{O}$), while p_e is the oracle's own failure rate, estimable by *validating* $\widehat{O}$ on points where c^* is known (measurement, or exact computation on tractable cases).

Theorem 1 (Candidate bound)**.** *For a fixed h, sample n observables $x_1, \ldots, x_n \sim D$ i.i.d. and let $\hat{p}_g$ be the empirical disagreement rate above $\varepsilon - \beta$. Then with probability $\geq 1 - \delta_s$,*

$$p_g(\varepsilon - \beta) \;\leq\; \hat{p}_g + \sqrt{\frac{\ln(1/\delta_s)}{2n}}.$$

This is a one-sided Hoeffding bound on a single proportion; no uniform-convergence assumption, no VC dimension.

Theorem 2 (Oracle validation bound)**.** *On m independent points $z_1, \ldots, z_m$ where $c^*(z)$ is known, let $\hat{p}_e$ be the empirical exceedance rate of $|\widehat{O} - c^*|$ above β. Then with probability $\geq 1 - \delta_e$,*

$$p_e(\beta) \;\leq\; \hat{p}_e + \sqrt{\frac{\ln(1/\delta_e)}{2m}}.$$

Theorem 3 (Two-level Valiant certificate)**.** *Fix $\varepsilon > 0$, $\beta \in (0, \varepsilon)$, and confidence budgets $\delta_s, \delta_e \in (0, 1)$. With probability at least $1 - \delta_s - \delta_e$,*

$$\mathrm{err}_D(h) \;\leq\; \hat{p}_g + \hat{p}_e + \sqrt{\frac{\ln(1/\delta_s)}{2n}} + \sqrt{\frac{\ln(1/\delta_e)}{2m}}.$$

The certificate is the tuple $(\varepsilon, \beta, n, m, \hat{p}_g, \hat{p}_e, \delta_s, \delta_e)$, from which the bound is computed. This is verification in the presence of an imperfect labeler—the generalization of PAC's clean oracle that string theory actually requires.

Remark 1 (The binding constraint)**.** *The oracle-validation radius is governed by m, the number of points where the true theory is known. For the Standard Model, m is bounded by the number of measured free parameters—not by compute. This is the crux of Section 5.*

4 Verification without learnability; search with it

Theorem 3 holds for a *single, handed-to-you* hypothesis h. It never invokes the complexity of the hypothesis class. That is the point: *verifying what you are given is cheap*, a Bernoulli-rate estimate.

Search is different. Search *selects* the empirically best of many hypotheses, and that selection is biased—the empirical minimum overestimates the winner. Bounding selection bias requires uniform convergence: for a finite class, a union bound gives radius $\sqrt{\ln(|C|/\delta)/(2n)}$; for an infinite class, $\mathrm{VCdim}(C)$ (or the statistical-query dimension $\mathrm{SQdim}(C)$) replaces $\ln|C|$.

The consequence is a graceful-degradation statement: if the learnability conjecture (Section 6) is false, the *search* is infeasible, but the *verifier survives*—it can still certify any candidate proposed externally, because the bound of Theorem 3 never uses VC dimension.

5 The observable measure and the measurable-physics bound

Make D *finite*: X is a finite, named list of N observables, each represented in log-space so that "tolerance ε" means a uniform *multiplicative* tolerance from a gauge coupling ($O(1)$) down to the cosmological constant (10^{-122}). D is then a weight vector $(w_1, \ldots, w_N)$.

For a finite X that the oracle can fully evaluate, the candidate term is *exact*:

$$p_g(\varepsilon - \beta) \;=\; \sum_{i=1}^{N} w_i \, \mathbf{1}\big[\, |\ln(h(x_i)/\widehat{O}(x_i))| > \varepsilon - \beta \,\big],$$

computed directly—no sampling, no n. The certificate collapses to the oracle term:

$$\mathrm{err}_D(h) \;\le\; p_g(\text{exact}) + \hat{p}_e + \sqrt{\frac{\ln(1/\delta_e)}{2m}}.$$

The only remaining uncertainty is the oracle-validation term, bounded by m, the number of measured points. With $m \sim 19$ (the Standard Model's free parameters) and $\delta_e = 0.025$:

$$\sqrt{\frac{\ln(40)}{2 \cdot 19}} \;\approx\; 0.31,$$

a $\sim$ 31–43% radius (the precise number depends on m and the confidence split). To drive the radius below 1% requires $m \approx 18{,}000$ measured points—*four orders of magnitude more than the theory gives us.* This is the quantitative statement of the verification deficit: string theory is today verifiable only to $\sim$ tens of percent tolerance in the observables that matter, and the bottleneck is measurable physics, not compute.

A second consequence is a 2×2 that decides which observables matter:

	c^* known (measured)	c^* unknown
$\widehat{O}$ can compute	verification zone	prediction zone
$\widehat{O}$ cannot compute	unverifiable	—

Only the top-left cell tightens the certificate; the top-right cell is falsifiable *later* and contributes nothing *now*. A vacuum whose only interesting predictions lie in the top-right cell is scientifically sterile *today*, regardless of elegance.

6 Search and the learnability conjecture

Valiant's evolvability model [5] restricts selection to *statistical-query* access to the target: fitness is read only through aggregate statistics, never per-instance ground truth. This restriction is not imposed here for tidiness—it is *already how the search works*, because the only access a search has to a vacuum's fitness is the certificate of Theorem 3 (sampled rates with confidence radii). Valiant's theorem is:

> *Evolvable* $\Rightarrow$ *learnable.* Any concept class evolvable in the restricted model is learnable in the corresponding statistical-query model.

Its contrapositive is the gate on search:

Conjecture 1 (Bounded SQ dimension)**.** *The class $C = \{h_v : X \to Y : v \in V\}$ of observable-prediction functions induced by vacua has* polynomial SQ dimension *—i.e., the compactification→observables map is SQ-learnable.*

- **True** $\Rightarrow$ the landscape is *evolvable*: an evolutionary search with certificate-only access and provable guarantees can find a near-optimal vacuum.
- **False** $\Rightarrow$ no search can find our vacuum—a negative result about the entire landscape program.

Either branch is a finding, and the conjecture is stated so it is attackable on small vacuum classes before anyone touches 10^{500}.

7 Discussion and limits

The framework does not unify. It does not select a vacuum—the measure (Section 1, claim 2) is untouched and remains the 40-year-open physics problem. It does not eliminate oracle error—it bounds it. It does not derive D—every certificate is conditional on the stipulated observable measure. And it does not turn a surrogate into truth: no error model, no certificate.

The imperfect oracle is the through-line. It appears in Theorem 3 (verification) and again in Section 6 (search, where fitness is read through certificates). A fully general treatment—*evolvability with an imperfect performance oracle*—is the natural extension and is left open.

What is buildable vs. not. All of Sections 3–6 is software (a reference implementation accompanies this paper). The measure problem, and the full EFT extraction that a non-trivial learnability measurement requires, are physics research, not tooling.

8 Conclusion

The obstacle to unified string theory is not a shortage of cleverness but a shortage of *checkable claims*—and the deepest checkable claim in reach is not "is this vacuum ours?" but "is the compactification→observables map learnable?" Answer that, and one knows whether the landscape is searchable at all. Verification, done with Valiant's guarantees, is the honest first move; selection is the hard second, and no amount of the first buys the second. The framework's contribution is to make the first move *rigorous* and to state, with a number, how far we are from being able to verify any of it.

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